

Q1: Attempt the following, all questions are compulsory, and each question carries 2 marks for each

- a) Prove by mathematical induction that $\sum_{i=1}^n \frac{1}{i(i+1)} = n/(n+1)$

Ans. We know that $\Rightarrow$

$$= \sum_{i=1}^n \frac{1}{i(i+1)} = n/n+1$$

By mathematical induction $\Rightarrow$

We have =

$$i=1$$

$$n=1$$

$$= \sum_{i=1}^1 \frac{1}{i(i+1)} = \frac{1}{2} \quad \text{--- (1)}$$

= Now $n = K, i = K$ Suppose $\Rightarrow$

$$= \sum_{i=1}^K \frac{1}{i(i+1)} = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{K(K+1)} \quad \text{--- (2)}$$

= then $n = K+1$

$$= \sum_{i=1}^{K+1} \frac{1}{(i)(i+1)}$$

$$= \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{K(K+1)} + \frac{1}{(K+1)(K+2)}$$

From eq (2) then

$$\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{-1}{K(K+1)} + \frac{1}{(K+1)(K+2)}$$

$$= \frac{K+2+K}{K(K+1)(K+2)} = \frac{2(K+1)}{K(K+1)(K+2)}$$

$$\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{n}{n+1}$$

Hence Proved

b) Verify whether $\sqrt{11}$ is rational or irrational.

Ans. The square root of 11, denoted as $\sqrt{11}$, is an irrational number.

To see why, suppose for the sake of contradiction that $\sqrt{11}$ is rational. This means that there exist two integers, say a and b , such that $\sqrt{11} = a/b$, where a and b have no common factors. Squaring both sides, we get:

$$11 = a^2 / b^2$$

Multiplying both sides by b^2 , we get:

$$11b^2 = a^2$$

This means that a^2 is divisible by 11, and hence a must also be divisible by 11 (since 11 is a prime number). Let $a = 11c$ for some integer c . Substituting this into the equation above, we get:

$$11b^2 = (11c)^2 = 121c^2$$

Dividing both sides by 11, we get:

$$b^2 = 11c^2$$

This means that b^2 is also divisible by 11, and hence b must also be divisible by 11. But this contradicts our assumption that a and b have no common factors, since both are divisible by 11. Therefore, our initial assumption that $\sqrt{11}$ is rational must be false, and hence $\sqrt{11}$ is irrational.

c) What are Demorgan's Law? Explain the use of Demorgan's law with example.

Ans. De Morgan's Laws: De Morgan's Laws are two fundamental principles in Boolean algebra that describe the relationship between the complement of a logical conjunction (AND) and the logical disjunction (OR), as well as the complement of a logical disjunction and the logical conjunction.

The first law states that the complement of a logical conjunction (the negation of the AND operator) is equivalent to the logical disjunction of the complements (the negation of each individual term) of the original expressions. Symbolically:

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$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

The second law states that the complement of a logical disjunction (the negation of the OR operator) is equivalent to the logical conjunction of the complements of the original expressions. Symbolically:

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

In simpler terms, De Morgan's Laws tell us that "not (A and B)" is the same as "(not A) or (not B)", and "not (A or B)" is the same as "(not A) and (not B)".

Use of Demorgan's law: De Morgan's Laws are useful in simplifying Boolean expressions and in reducing the number of gates needed to implement a logical circuit.

Example : An example of using De Morgan's Laws is:

Suppose we want to simplify the Boolean expression $\neg(p \wedge \neg q)$.

Using the first law of De Morgan, we can write this as:

$$\neg(p \wedge \neg q) \equiv \neg p \vee q$$

Similarly, if we want to simplify the Boolean expression $\neg(p \vee \neg q)$, we can use the second law of De Morgan to write:

$$\neg(p \vee \neg q) \equiv \neg p \wedge q$$

d) Make truth table for followings:

i) $p \rightarrow (\neg q \vee \neg r) \wedge (p \vee r)$

Ans. Here is the truth table:

p	q	r	$\neg q \vee \neg r$	$p \vee r$	$(p \rightarrow (\neg q \vee \neg r) \wedge (p \vee r))$
T	T	T	F	T	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	F	F

ii) $p \rightarrow (\neg r \wedge q) \wedge (p \wedge \neg q)$

Ans. Here is the truth table:

p	q	r	$\neg r \wedge q$	$p \wedge \neg q$	$(p \rightarrow (\neg r \wedge q) \wedge (p \wedge \neg q))$
T	T	T	F	F	F
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	F	F	T
F	T	F	F	F	T
F	F	T	F	F	T
F	F	F	F	F	T

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e) Obtain the truth value of disjunction of "Water is essential for life" and " $2+2=4$ ".

Ans. The truth value of a disjunction (logical OR) is true if at least one of the statements being connected is true.

In this case, "Water is essential for life" is a true statement, and " $2+2=4$ " is also a true statement. Therefore, the disjunction of these statements is true.

So, the truth value of "Water is essential for life" OR " $2+2=4$ " is true.

f) Write the following statements in the symbolic form.

i) Some students can not appear in exam.

Ans. Let $S(x)$ be the statement "x is a student" and $E(x)$ be the statement "x can appear in the exam". Then the statement "Some students can not appear in exam" can be written symbolically as:

$$\exists x (S(x) \wedge \neg E(x))$$

This reads as "there exists an x such that x is a student and x cannot appear in the exam".

ii) Everyone can not sing.

Ans. Let $P(x)$ be the statement "x is a person" and $S(x)$ be the statement "x can sing". Then the statement "Everyone can not sing" can be written symbolically as:

$$\neg \forall x (P(x) \rightarrow S(x))$$

This reads as "it is not the case that for all x, if x is a person then x can sing". Alternatively, we can also write this statement as:

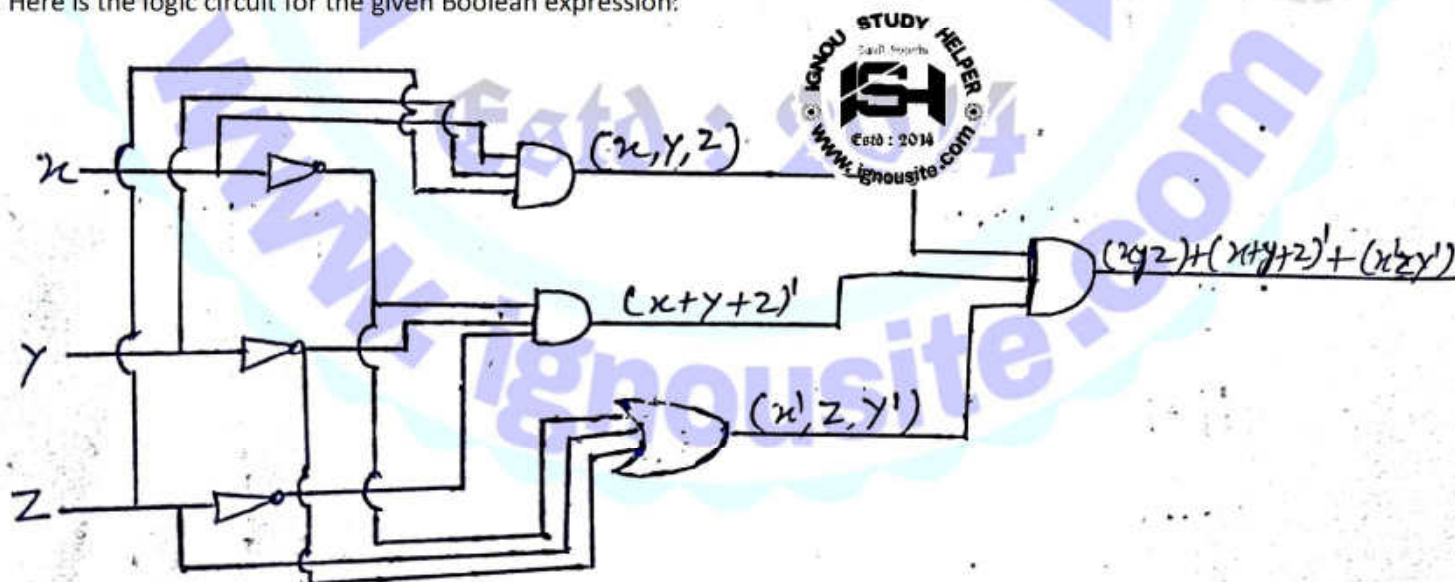
$$\exists x (P(x) \wedge \neg S(x))$$

This reads as "there exists an x such that x is a person and x cannot sing".

g) Draw logic circuit for the following Boolean Expression:

$$(x y z) + (x+y+z)' + (x'zy')$$

Ans. Here is the logic circuit for the given Boolean expression:



h) What is dual of a boolean expression? Explain with the help of an example.

Ans. Dual of a boolean expression: In Boolean algebra, the dual of an expression is obtained by interchanging the operators AND and OR, and replacing 0's with 1's and 1's with 0's. In other words, if we have a Boolean expression in terms of AND, OR, and NOT operators, the dual of that expression is obtained by replacing all instances of AND with OR, OR with AND, 0 with 1, and 1 with 0.

Example: For example, let's consider the Boolean expression:

$A \text{ AND } (B \text{ OR } C)$

To find its dual, we first interchange the AND and OR operators:

$A \text{ OR } (B \text{ AND } C)$

Next, we replace 0's with 1's and 1's with 0's:

$\text{NOT } A \text{ AND } (\text{NOT } B \text{ AND } \text{NOT } C)$

So the dual of the original expression " $A \text{ AND } (B \text{ OR } C)$ " is " $\text{NOT } A \text{ AND } (\text{NOT } B \text{ AND } \text{NOT } C)$ ".

i) Show using truth table whether $(P \wedge Q \vee R)$ and $(P \vee R) \wedge (Q \vee R)$ are equivalent or not.

Ans. To determine whether the two expressions " $(P \wedge Q \vee R)$ " and " $(P \vee R) \wedge (Q \vee R)$ " are equivalent or not, we can construct a truth table for both expressions and compare their output for all possible combinations of truth values of P, Q, and R.

Here is the truth table for " $(P \wedge Q \vee R)$ ":

P	Q	R	$Q \vee R$	$P \wedge Q \vee R$
0	0	0	0	0
0	0	1	1	1
0	1	0	1	0
0	1	1	1	1
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

And here is the truth table for " $(P \vee R) \wedge (Q \vee R)$ ":

P	Q	R	$Q \vee R$	$P \vee R$	$(P \vee R) \wedge (Q \vee R)$
0	0	0	0	0	0
0	0	1	1	1	1
0	1	0	1	0	0
0	1	1	1	1	1
1	0	0	1	1	0
1	0	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1

Now, we can compare the output of both expressions for each combination of P, Q, and R. By looking at the truth tables, we can see that the two expressions have the same output for all possible combinations of truth values of P, Q, and R. Therefore, we can conclude that " $(P \wedge Q \vee R)$ " and " $(P \vee R) \wedge (Q \vee R)$ " are equivalent.

j) Explain whether $(P \wedge Q) \rightarrow (Q \rightarrow R)$ is a tautology or not.

Ans. To determine whether the proposition " $(P \wedge Q) \rightarrow (Q \rightarrow R)$ " is a tautology or not, we need to construct a truth table for all possible combinations of truth values of P, Q, and R, and evaluate the proposition for each combination.

Here is the truth table for " $(P \wedge Q) \rightarrow (Q \rightarrow R)$ ":

P	Q	R	$P \wedge Q$	$Q \rightarrow R$	$(P \wedge Q) \rightarrow (Q \rightarrow R)$
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0	0	0	1
0	1	1	0	1	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	0	0
1	1	1	1	1	1

As we can see from the truth table, the proposition is not a tautology because there is at least one combination of truth values where the proposition is false (when P is true, Q is true, and R is false).

Therefore, we can conclude that " $(P \wedge Q) \rightarrow (Q \rightarrow R)$ " is not a tautology.

Q2: Attempt the following, all questions are compulsory, and each question carries 2 marks for each

a) Set A, B and C are: $A = \{1, 2, 3, 5, 7, 9, 11, 13\}$, $B = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $C = \{1, 2, 4, 5, 6, 7, 8, 10, 13\}$. Find $A \cap B \cap C$, $A \cup B \cup C$, $A \cup B \cap C$

Ans. To solve this problem, we need to understand the set operations of intersection ($\cap$) and union ($\cup$).

- The intersection ($\cap$) of two sets A and B is the set of elements that are common to both sets. Symbolically, $A \cap B = \{x : x \in A \text{ and } x \in B\}$.
- The union ($\cup$) of two sets A and B is the set of all elements that are in either set. Symbolically, $A \cup B = \{x : x \in A \text{ or } x \in B\}$.

Now, let's use these definitions to solve the problem:

$$A = \{1, 2, 3, 5, 7, 9, 11, 13\} \quad B = \{1, 2, 3, 4, 5, 6, 7, 8, 9\} \quad C = \{1, 2, 4, 5, 6, 7, 8, 10, 13\}$$

$A \cap B \cap C$ is the intersection of all three sets, which is the set of elements that are in all three sets:

$$A \cap B \cap C = \{1, 2, 5, 7\}$$

$A \cup B$ is the union of sets A and B, which is the set of elements that are in either set:

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 13\}$$

$A \cup B \cup C$ is the union of all three sets, which is the set of elements that are in any of the sets:

$$A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13\}$$

$A \cup B \cap C$ is the union of sets A and B, but only with the elements that are also in set C. In other words, it is the set of elements that are in either A or B, but not both, and are also in C:

$$A \cup B \cap C = \{1, 2, 5, 7\} \text{ (because 4, 6, 8, 9, 10, 11, and 13 are not in A or B).}$$

b) What is power set? Write power set of set $A=\{1, 2, 3, 4, 5, 6, 7, 9\}$.

Ans. Power set: The power set of a set A , denoted by $P(A)$, is the set of all possible subsets of A , including the empty set and A itself. In other words, the power set of A is the set of all possible combinations of elements from A .

To find the power set of $A=\{1, 2, 3, 4, 5, 6, 7, 9\}$, we can use the following steps:

1. Count the number of elements in A . In this case, there are 8 elements.
2. Determine the number of subsets of A by using the formula 2^n , where n is the number of elements in A . In this case, $2^8 = 256$.
3. List all possible subsets of A by including or excluding each element. There are 256 subsets, including the empty set and the set A itself.

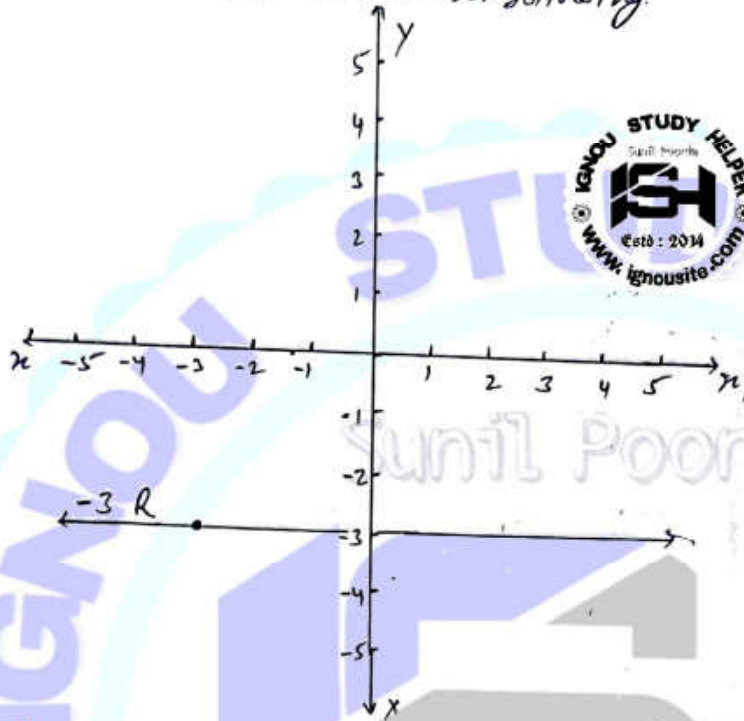
The power set of A is:

$\{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}, \{7\}, \{9\},$
 $\{1,2\}, \{1,3\}, \{1,4\}, \{1,5\}, \{1,6\}, \{1,7\}, \{1,9\},$
 $\{2,3\}, \{2,4\}, \{2,5\}, \{2,6\}, \{2,7\}, \{2,9\}, \{3,4\},$
 $\{3,5\}, \{3,6\}, \{3,7\}, \{3,9\},$
 $\{4,5\}, \{4,6\}, \{4,7\}, \{4,9\},$
 $\{5,6\}, \{5,7\}, \{5,9\},$
 $\{6,7\}, \{6,9\},$
 $\{7,9\},$
 $\{1,2,3\}, \{1,2,4\}, \{1,2,5\}, \{1,2,6\}, \{1,2,7\}, \{1,2,9\},$
 $\{1,3,4\}, \{1,3,5\}, \{1,3,6\}, \{1,3,7\}, \{1,3,9\},$
 $\{1,4,5\}, \{1,4,6\}, \{1,4,7\}, \{1,4,9\},$
 $\{1,5,6\}, \{1,5,7\}, \{1,5,9\},$
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 $\{2,6,7\}, \{2,6,9\},$
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 $\{4,7,9\},$
 $\{5,6,7\}, \{5,6,9\},$
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 $\{6,7,9\},$
 $\{7,9\}$
 $\{1,2,3,4,5,6,7,9\}$

c) Give geometric representation for followings:

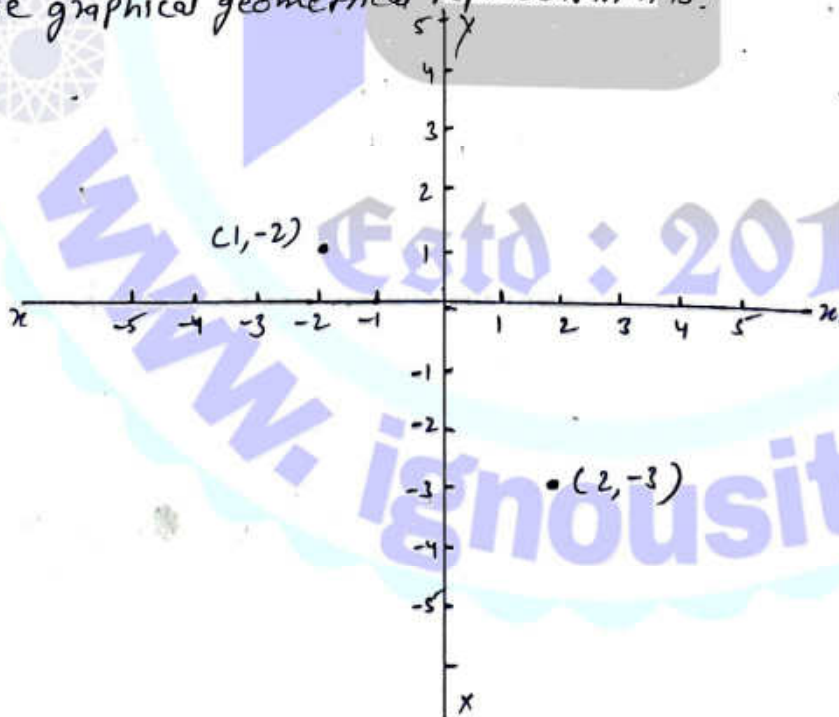
i) $\{-3\} \times \mathbb{R}$

Ans. $\{-3\} \times \mathbb{R}$ geometric representation for following.



ii) $\{1, -2\} \times \{2, -3\}$

Ans. $\{1, -2\} \times \{2, -3\} \Rightarrow$ we have $\{1, -2\} \times \{2, -3\} \Rightarrow \{(1, -2), (2, -3)\}$
whose graphical geometrical representation is!



d) What is proper subset? Explain with the help of example.

Ans. Proper subset: A proper subset is a subset that contains only a part of the elements of the original set, excluding at least one element that is not in the subset. In other words, if all the elements of a set A are also in another set B, and B contains at least one element that is not in A, then A is a proper subset of B.

We can represent the relationship between sets using set notation. If A is a proper subset of B, we write $A \subset B$ or $B \supset A$. The symbol $\subset$ means "is a proper subset of" or "is contained in but not equal to."

Example: For example, let's consider two sets: $A = \{1, 2, 3\}$ $B = \{1, 2, 3, 4\}$

Here, set A is a proper subset of set B because every element of A is also in B, but B contains an additional element (4) that is not in A. We can represent this as $A \subset B$ or $B \supset A$.

On the other hand, if we have two sets with the same elements, they are not considered proper subsets of each other. For example, if $C = \{1, 2, 3\}$, then C is not a proper subset of A, and A is not a proper subset of C, because they have the same elements.

e) What is relation? Explain properties of relations with example.

Ans. Relation: In mathematics, a relation is a connection or association between two or more things. In set theory, a relation between two sets is a set of ordered pairs, where each ordered pair contains one element from each set. A relation can also be thought of as a rule or condition that defines how the elements of one set relate to the elements of another set.

There are several properties that relations can have. Here are some common ones:

1. **Reflexive:** A relation R on a set A is reflexive if for every element $a \in A$, $(a,a) \in R$. In other words, every element of A is related to itself.
Example: For example, the relation "is equal to" is reflexive, since every element is equal to itself.
2. **Symmetric:** A relation R on a set A is symmetric if for every pair of elements $(a,b) \in R$, $(b,a) \in R$. In other words, if a is related to b, then b is related to a.
Example: For example, the relation "is a sibling of" is symmetric, since if x is a sibling of y, then y is a sibling of x.
3. **Transitive:** A relation R on a set A is transitive if for every three elements a, b, and c $\in A$, if $(a,b) \in R$ and $(b,c) \in R$, then $(a,c) \in R$. In other words, if a is related to b and b is related to c, then a is related to c.
Example: For example, the relation "is an ancestor of" is transitive, since if x is an ancestor of y and y is an ancestor of z, then x is an ancestor of z.
4. **Antisymmetric:** A relation R on a set A is antisymmetric if for every pair of distinct elements $(a,b) \in R$, $(b,a) \notin R$. In other words, if a is related to b, then b is not related to a.
Example: For example, the relation "is less than" is antisymmetric, since if x is less than y and y is less than x, then x and y must be equal.

f) Explain whether function: $f(x) = x^2$ possesses an inverse function or not.

Ans. The function $f(x) = x^2$ does not have an inverse function over the set of real numbers.

To show this, we need to demonstrate that the function is not one-to-one, meaning that two different inputs can produce the same output. In this case, for any given positive value of x, both x and -x produce the same output value of x^2 . For example, $f(2) = 4$ and $f(-2) = 4$, so the outputs 4 have two distinct inputs, namely 2 and -2. Therefore, $f(x) = x^2$ is not one-to-one, and thus does not have an inverse function over the set of real numbers.

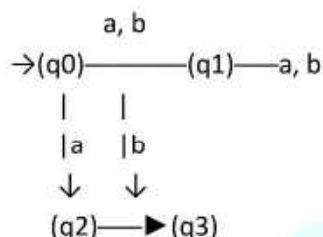
However, if we restrict the domain of the function to only non-negative values of x, then it does have an inverse function, which is $f^{-1}(x) = \sqrt{x}$. In this case, the function is one-to-one and each output has a unique input.

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g) Write the finite automata corresponding to the regular expression $(a + b)^*ab$

Ans. The regular expression $(a + b)^*ab$ represents the set of all strings that end with the string "ab".

Here is the finite automaton that corresponds to this regular expression:



In this automaton, q_0 is the start state, and q_3 is the accepting state. The arrows represent transitions on input symbols.

From state q_0 , the automaton can transition to either q_0 or q_1 on input symbol "a" or "b". From q_1 , the automaton can only transition to q_2 on input symbol "a" and to q_3 on input symbol "b". Finally, from q_2 , the automaton can only transition to q_3 on input symbol "b".

To recognize a string using this automaton, we start at the start state q_0 and read the input symbols one by one. If we reach the end of the string and we are in the accepting state q_3 , then the string is accepted by the automaton and is in the language represented by the regular expression $(a + b)^*ab$.

h) If L_1 and L_2 are context free languages then, prove that $L_1 \cup L_2$ is a context free language.

Ans. To prove that $L_1 \cup L_2$ is a context-free language, we can construct a context-free grammar that generates the language. Let $G_1 = (V_1, \Sigma, R_1, S_1)$ be a context-free grammar for L_1 and $G_2 = (V_2, \Sigma, R_2, S_2)$ be a context-free grammar for L_2 .

We can construct a new grammar $G = (V, \Sigma, R, S)$ for $L_1 \cup L_2$ by introducing a new start symbol S and adding the productions:

$$S \rightarrow S_1 \mid S_2$$

The non-terminal symbols V of the new grammar G are the union of the non-terminal symbols of G_1 and G_2 , i.e. $V = V_1 \cup V_2 \cup \{S\}$. The set of terminals Σ is the same for both languages L_1 and L_2 , and R is the union of R_1 and R_2 together with the new production $S \rightarrow S_1 \mid S_2$.

The production $S \rightarrow S_1$ generates all strings in L_1 , and $S \rightarrow S_2$ generates all strings in L_2 . Since G_1 and G_2 are context-free, the productions in R_1 and R_2 are context-free, and therefore the productions in R are also context-free.

Thus, we have shown that $L_1 \cup L_2$ is a context-free language.

i) Explain Decidable and Undecidable Problems. Give example for each.

Ans. Decidable Problem: A decision problem is decidable if there exists an algorithm that can solve it in a finite amount of time for any given input. In other words, a decidable problem is one for which there is a definitive and algorithmic way to determine the answer.

Example: Given a list of numbers, is there a pair of numbers in the list whose sum is equal to a specific value? This is a decidable problem because we can solve it using a straightforward algorithm. We can iterate over all possible pairs of numbers in the list and check whether their sum equals the specific value. If we find such a pair, we can return "yes"; otherwise, we can return "no".

Undecidable Problem: An undecidable problem is one for which there is no algorithm that can solve it for all possible inputs. That is, there is no finite algorithm that can always provide a definite yes or no answer to the problem.

Example: The halting problem is a classic example of an undecidable problem. Given a program and an input, the halting problem asks whether the program will eventually halt (i.e., stop running) when given that input. It has been proven that there is no algorithm that can solve the halting problem for all possible programs and inputs. This means that there are some programs for which it is impossible to determine whether they will halt or run indefinitely.

j) What is equivalence relation? Explain use of equivalence relation with the help of an example.

Ans. Equivalence relation: An equivalence relation is a relation between elements of a set that satisfies three conditions: reflexivity, symmetry, and transitivity. That is, if we have a set S and a relation R on S , then R is an equivalence relation if and only if:

1. Reflexivity: for any element a in S , aRa (that is, a is related to itself).
2. Symmetry: for any elements a and b in S , if aRb , then bRa (that is, if a is related to b , then b is related to a).
3. Transitivity: for any elements a , b , and c in S , if aRb and bRc , then aRc (that is, if a is related to b and b is related to c , then a is related to c).

Use of equivalence relation: Equivalence relations are useful in many areas of mathematics and computer science, including group theory, graph theory, and relational databases.

Example: One example of an equivalence relation is the relation of congruence modulo n on the set of integers Z . For any integers a and b , we say that a is congruent to b modulo n (written $a \equiv b \pmod{n}$) if and only if $a - b$ is divisible by n . This relation is an equivalence relation because it satisfies the three conditions:

1. Reflexivity: for any integer a , $a \equiv a \pmod{n}$, because $a - a = 0$ is divisible by n .
2. Symmetry: if $a \equiv b \pmod{n}$, then $a - b$ is divisible by n . But if $a - b$ is divisible by n , then so is $-(a - b) = b - a$, so $b \equiv a \pmod{n}$.
3. Transitivity: if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a - b$ is divisible by n and $b - c$ is divisible by n . Therefore, $(a - b) + (b - c) = a - c$ is divisible by n , so $a \equiv c \pmod{n}$.

This relation is useful in many applications, including cryptography and number theory.

Q3: Attempt the following, all questions are compulsory, and each question carries 2 marks for each.

a) Suppose we want to choose two persons from a party consisting of 35 members as Manager and Assistant Manager. In how many ways can this be done?

Ans. The number of ways to choose two persons out of 35 is given by the combination formula:

$$C(35, 2) = 35! / (2! * (35-2)!) = (35 * 34) / (2 * 1) = 595$$

Therefore, there are 595 ways to choose two persons from a party consisting of 35 members to be the Manager and Assistant Manager.

b) There are three Companies, C1, C2 and C3. The party C1 has 4 members, C2 has 5 members and C3 has 6 members in an assembly. Suppose we want to select two persons, both from the same Company, to become president and vice president. In how many ways can this be done?

Ans. We need to select two persons from the same company and then select one as president and the other as vice president. There are three companies, and for each company, we need to choose 2 persons, and then we need to select one person as president and the other as vice president. Thus, the total number of ways is:

$$3 * C(4, 2) * 2! + 3 * C(5, 2) * 2! + 3 * C(6, 2) * 2!$$

where $C(n, k)$ is the number of ways to choose k items from a set of n items.

Evaluating this expression, we get:

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$$3 * C(4, 2) * 2! + 3 * C(5, 2) * 2! + 3 * C(6, 2) * 2!$$

$$= 3 * 6 * 2 + 3 * 10 * 2 + 3 * 15 * 2$$

$$= 36 + 60 + 90$$

$$= 186$$

Therefore, there are 186 ways to select two persons, both from the same Company, to become president and vice president.

c) Suppose there are five married couples and they (10 people) are made to sit about a round table so that neither two men nor two women sit together. Find the number of such circular arrangements.

Ans. Since the 10 people are seated in a circular arrangement, we can fix the position of one person and arrange the others relative to that person. Let's choose one of the 5 husbands to be in a fixed position. We can then arrange the remaining 9 people in a line, such that no two men or two women are seated together.

First, we arrange the 5 wives in a line, such that no two women are seated together. This can be done in 4! ways.

Next, we insert the 5 husbands into the gaps between the wives, such that no two men are seated together. We can think of the 4 gaps between the wives as "slots" for the husbands. There are 4! ways to arrange the slots with no restrictions. However, we need to remove the arrangements in which two or more husbands are seated together. If two husbands are seated together, we can treat them as a single entity and arrange the remaining entities in 3! ways (since there are 3 entities remaining: 3 wives, 1 group of two husbands). If three or more husbands are seated together, we can treat them as a single entity and arrange the remaining entities in 2! ways (since there are 2 entities remaining: 2 wives, 1 group of three or more husbands).

Therefore, the total number of circular arrangements is:

$$4! \times (4! - 2 \times 3! + 2 \times 2!)$$

$$= 24 \times (24 - 12 + 4)$$

$$= 24 \times 16$$

$$= 384$$

Thus, there are 384 possible circular arrangements of the 10 people, such that no two men or two women are seated together.

d) How many words can be formed using letter of DEPARTMENT using each letter at most once?

i) If each letter must be used,

Ans. In the word "DEPARTMENT", total number of letters = 10 among which 'E' appears 2 times, 'T' appears 2 times and other than 'E' and 'T' all letters appearing 1 time each.

So the number of distinct letters = $(10 - 2) = 7$ (D, E, P, A, R, T, M, N)

The number of words can be formed using all the distinct letters of the word "DEPARTMENT" is given by –

$$= 8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40,320$$

ii) If some or all the letters may be omitted.

Ans. If some or all the letters may be omitted, we can form words of different lengths. We can form words of length 1, 2, 3, ..., up to 10 by selecting a subset of the letters in DEPARTMENT and arranging them in all possible orders.

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For a word of length k , we need to select k letters from DEPARTMENT and arrange them in $k!$ ways. The total number of words that can be formed is the sum of the number of words of length 1, 2, 3, ..., up to 10.

Number of words of length 1 = 10

Number of words of length 2 = $C(10, 2) \times 2! = 45 \times 2 = 90$

Number of words of length 3 = $C(10, 3) \times 3! = 120 \times 6 = 720$

Number of words of length 4 = $C(10, 4) \times 4! = 210 \times 24 = 5,040$

Number of words of length 5 = $C(10, 5) \times 5! = 252 \times 120 = 30,240$

Number of words of length 6 = $C(10, 6) \times 6! = 210 \times 720 = 151,200$

Number of words of length 7 = $C(10, 7) \times 7! = 120 \times 5,040 = 604,800$

Number of words of length 8 = $C(10, 8) \times 8! = 45 \times 40,320 = 1,814,400$

Number of words of length 9 = $C(10, 9) \times 9! = 10 \times 362,880 = 3,628,800$

Number of words of length 10 = $C(10, 10) \times 10! = 1 \times 3,628,800 = 3,628,800$

The total number of words that can be formed by using some or all the letters in DEPARTMENT is the sum of the above numbers, which is:

$$10 + 90 + 720 + 5,040 + 30,240 + 151,200 + 604,800 + 1,814,400 + 3,628,800 + 3,628,800 = 10,393,980$$

Therefore, there are 10,393,980 words that can be formed using the letters of DEPARTMENT, where each letter may be used at most once.

e) What is the probability that a 13-card hand has at least one card in each suit?

Ans. The probability that a 13-card hand does not have at least one card in each suit is equal to the sum of the probabilities that the hand has no cards from one or more of the four suits.

The probability that a 13-card hand has no cards from a particular suit is given by the number of ways to choose 13 cards from the remaining 39 cards of the other three suits divided by the total number of ways to choose 13 cards from the full deck of 52 cards:

$$P(\text{No cards in a particular suit}) = C(39, 13) / C(52, 13)$$

Since there are four suits, the probability that a 13-card hand does not have at least one card in each suit is:

$$P(\text{No cards in all suits}) = 4 \times P(\text{No cards in a particular suit}) = 4 \times C(39, 13) / C(52, 13)$$

Therefore, the probability that a 13-card hand has at least one card in each suit is:

$$P(\text{At least one card in each suit}) = 1 - P(\text{No cards in all suits}) = 1 - 4 \times C(39, 13) / C(52, 13) \approx 0.724$$

Thus, the probability that a 13-card hand has at least one card in each suit is approximately 0.724.

f) What is the probability that a number between 1 and 10,000 is divisible by neither 2, 3, 5 nor 7?

Ans. Let N be the total number of integers between 1 and 10,000, inclusive. We can count the integers between 1 and 10,000 that are divisible by 2, 3, 5, or 7 and use the principle of inclusion-exclusion to find the number of integers that are divisible by at least one of these numbers. Then, we can subtract this count from N to get the number of integers that are not divisible by any of them. Finally, we divide this number by N to get the probability that a number between 1 and 10,000 is not divisible by 2, 3, 5, or 7.

There are 5,000 even integers between 1 and 10,000. There are 3,333 integers divisible by 3, 2,000 integers divisible by 5, and 1,428 integers divisible by 7. We need to subtract the integers that are divisible by at least two of these numbers, then add back the integers that are divisible by at least three of them, and finally subtract the integers that are divisible by all four of them. To do this, we use the principle of inclusion-exclusion:

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Number of integers divisible by at least one of 2, 3, 5, or 7 = $5000 + 3333 + 2000 + 1428 - (1666 + 1000 + 714 + 285) + (333 + 142 + 95) - 47 = 7680$

Therefore, the number of integers between 1 and 10,000 that are not divisible by 2, 3, 5, or 7 is:

$$N - 7680 = 10,000 - 7680 = 2,320$$

The probability that a number between 1 and 10,000 is not divisible by 2, 3, 5, or 7 is:

$$P(\text{not divisible by 2, 3, 5, or 7}) = 2320 / 10000 = 0.232$$

Thus, the probability that a number between 1 and 10,000 is not divisible by 2, 3, 5, or 7 is 0.232.

g) Explain inclusion-exclusion principle and Pigeon Hole Principle with example.

Ans. Inclusion-Exclusion Principle: The inclusion-exclusion principle is a counting technique used to find the size of the union of two or more sets. It states that the size of the union of two or more sets is equal to the sum of their sizes, minus the size of their intersection(s), plus the size of the intersection(s) of their pairwise combinations, minus the size of the intersection(s) of their triple-wise combinations, and so on. The formula for the inclusion-exclusion principle is as follows:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

and so on for any number of sets.

Example: Suppose we have three sets A, B, and C, where $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7\}$. We want to find the size of their union.

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C| = 4 + 4 + 4 - 2 - 1 - 1 + 1 = 9$$

Therefore, the size of the union of A, B, and C is 9.

Pigeonhole Principle: The pigeonhole principle is a basic principle in combinatorics that states that if n items are placed into m containers where $n > m$, then at least one container must contain more than one item. The principle is often used to show that a certain situation or outcome is inevitable or to find lower bounds on the number of objects that must be used to satisfy a certain property.

Example: Suppose we have 10 apples and 9 baskets. We want to put the apples into the baskets such that each basket has at most one apple. Since we have more apples than baskets, by the pigeonhole principle, there must be at least one basket that has more than one apple.

h) In a tennis tournament, each entrant plays a match in the first round. Next, all winners from the first round play a second-round match. Winners continue to move on to the next round, until finally only one player is left as the tournament winner. Assuming that tournaments always involve $n = 2^k$ players, for some k , find the recurrence relation for the number rounds in a tournaments of n players.

Ans. Let $T(n)$ denote the number of rounds required for a tournament with n players.

In the first round, each player plays exactly one match, and there are $n/2$ matches. The winners of each of these matches move on to the second round, so there are $n/2$ players in the second round. Since the number of players is now halved, the number of rounds required to complete the second round is $T(n/2)$. Similarly, the winners of the second round move on to the third round, so there are $n/4$ players in the third round, and the number of rounds required to complete the third round is $T(n/4)$. This continues until there are only two players left, who play in the final round.

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Therefore, we can write the recurrence relation for $T(n)$ as:

$$T(n) = T(n/2) + 1 \text{ for } n > 2 \quad T(2) = 1$$

The base case is $T(2) = 1$, since there is only one match in this case.

This recurrence relation can be solved using the master theorem, which gives the solution as $T(n) = \log_2 n$. Therefore, a tournament with n players requires $\log_2 n$ rounds.

i) Find an explicit recurrence relation for minimum number of moves in which the n -disks in tower of Hanoi puzzle can be solved! Also solve the obtained recurrence relation through an iterative method.

Ans. The minimum number of moves required to solve the Tower of Hanoi puzzle with n disks can be obtained recursively as follows:

Let $M(n)$ denote the minimum number of moves required to solve the puzzle with n disks.

- If $n=1$, it can be solved in 1 move.
- If $n>1$, then we need to move the top $(n-1)$ disks to a spare peg, move the n th disk to the target peg, and then move the $(n-1)$ disks from the spare peg to the target peg. This requires $2M(n-1) + 1$ moves.

Therefore, we have the following recurrence relation for the minimum number of moves:

$$M(n) = 1 \text{ if } n=1 \quad M(n) = 2M(n-1) + 1 \text{ if } n>1$$

Using this recurrence relation, we can solve for $M(n)$ iteratively as follows:

$$\begin{aligned} M(1) &= 1 \\ M(2) &= 2M(1) + 1 = 3 \\ M(3) &= 2M(2) + 1 = 7 \\ M(4) &= 2M(3) + 1 = 15 \\ M(5) &= 2M(4) + 1 = 31 \dots \end{aligned}$$



In general, we can obtain $M(n)$ by doubling the previous value of M and adding 1. Therefore, the explicit formula for $M(n)$ is:

$$M(n) = 2^n - 1$$

So, the minimum number of moves required to solve the Tower of Hanoi puzzle with n disks is $2^n - 1$.

j) Find the solution of the recurrences relation $a_n = a_{n-1} + 2a_{n-2}$, $n > 2$ with $a_0 = 0$, $a_1 = 1$

Ans. Given the recurrence relation:

$$a_n = a_{n-1} + 2a_{n-2}, \quad n > 2 \text{ with } a_0 = 0, a_1 = 1$$

To solve this recurrence relation, we can use the characteristic equation method. We assume that the solution has the form:

$$a_n = r^n$$

Substituting this into the recurrence relation, we get:

$$r^n = r^{(n-1)} + 2r^{(n-2)}$$

Dividing both sides by $r^{(n-2)}$, we get:

$$r^2 - r - 2 = 0$$

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Solving this quadratic equation, we get:

$$r = 2 \text{ or } r = -1$$

So, the general solution of the recurrence relation is given by:

$$a_n = A(2^n) + B(-1)^n$$

where A and B are constants that can be determined from the initial conditions:

$$a_0 = 0 \rightarrow A + B = 0 \quad a_1 = 1 \rightarrow 2A - B = 1$$

Solving these equations simultaneously, we get:

$$A = 1/3 \text{ and } B = -1/3$$

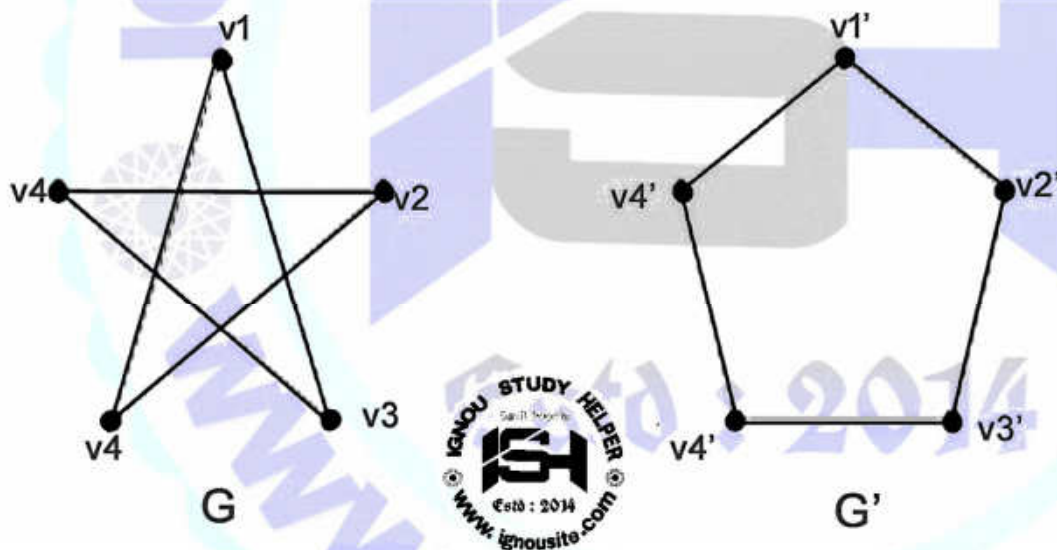
Therefore, the solution of the recurrence relation is:

$$a_n = (1/3)(2^n) - (1/3)(-1)^n$$

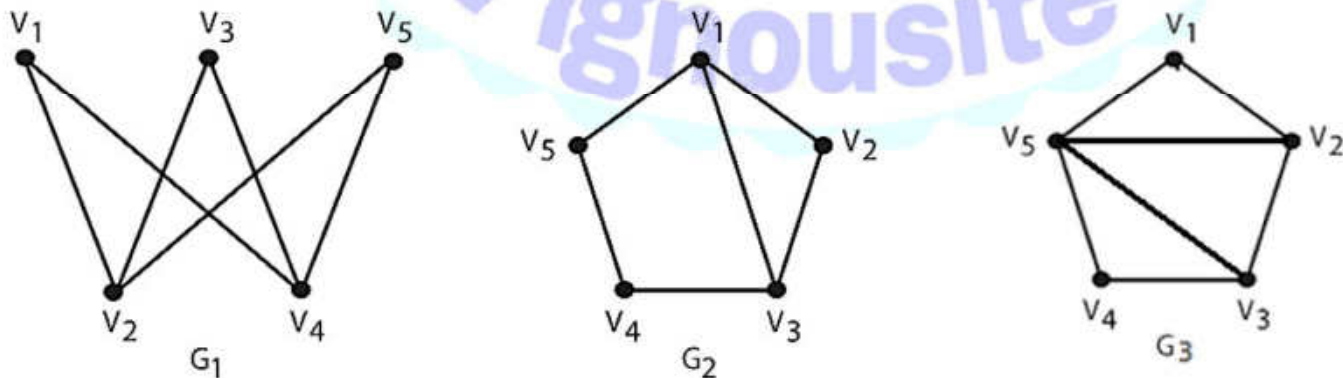
Q4: Attempt the following, all questions are compulsory, and each question carries 2 marks for each

a) Draw 2-isomorphic graphs and 3 non-isomorphic graphs on five vertices.

Ans. 2-isomorphic graphs on five vertices: -



3 non-isomorphic graphs on five vertices: -



b) Prove that the complement of $\bar{G}$ is G .

Ans. The complement of a graph G is another graph that has the same set of vertices as G , but has edges between any two vertices in G that are not adjacent in G , and does not have edges between any two vertices in G that are adjacent in G .

To prove that the complement of G is G , we need to show that every vertex in G is adjacent to every vertex that is not adjacent to it in G , and that no vertex in G is adjacent to any vertex that is adjacent to it in G .

Let u and v be two vertices in G that are not adjacent in G . Then in the complement of G , there is an edge between u and v , since they are not adjacent in G . But this means that in G , u and v are not connected by an edge. Similarly, if u and v are adjacent in G , then in the complement of G , there is no edge between u and v , since they are adjacent in G . This means that in G , u and v are connected by an edge.

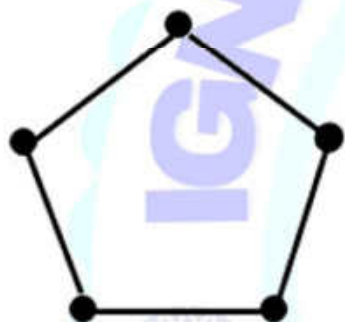
Since this holds for every pair of vertices in G , we can conclude that the complement of $\bar{G}$ is G .

c) Draw the following graphs and state which of following graph is a regular graph?

- (i) C_5 (ii) W_5 (iii) Q_4 (iv) $K_{5,5}$

Ans.

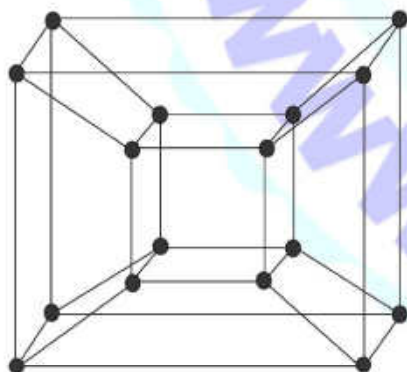
(i) C_5



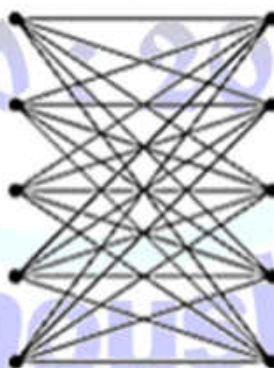
(ii) W_5



(iii) Q_4



(iv) $K_{5,5}$



A graph is regular if all its vertices have the same degree.

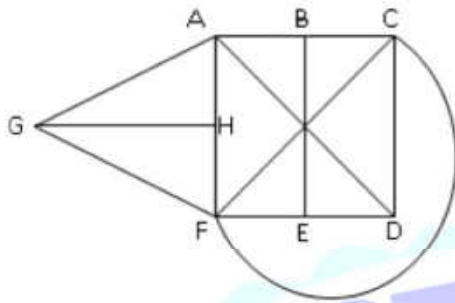
(i) C_5 is a regular graph because each vertex has degree 2.

(ii) W_5 is not a regular graph because the center vertex has degree 4 while the others have degree 3.

(iii) Q_4 is a regular graph because each vertex has degree 3.

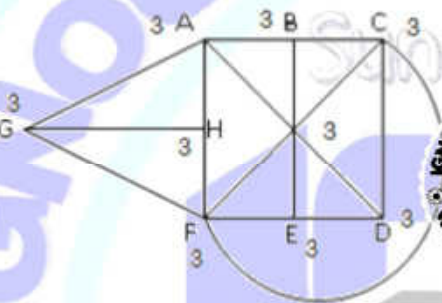
(iv) $K_{5,5}$ is not a regular graph because each vertex on the left has degree 5 while each vertex on the right has degree 5.

d) What is a chromatic number of a graph? What is a chromatic number of the following graph?



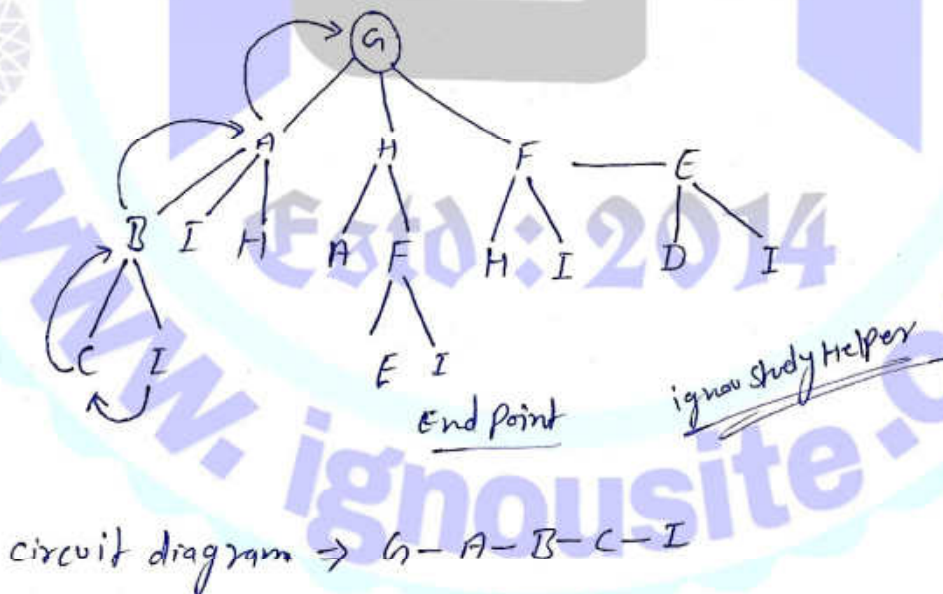
Ans. Chromatic number of a graph: The chromatic number of a graph is the minimum number of colors needed to color the vertices of the graph so that no adjacent vertices share the same color. In other words, it is the minimum number of colors needed to color the vertices of the graph in such a way that no two adjacent vertices have the same color.

Chromatic number: The chromatic number of this graph is 3.



e) Determine whether the above graph has a Hamiltonian circuit. If it has, find such a circuit. If it does not have, justify it

Ans. The above graph is Hamiltonian circuit:



f) Explain and prove the Handshaking Theorem, with suitable example

Ans. Handshaking Theorem: The Handshaking Theorem, also known as the Handshaking Lemma or the Handshake Lemma, is a result in graph theory that provides a relationship between the number of edges and the degrees of vertices in an undirected graph. The theorem states that the sum of the degrees of all vertices in an undirected graph is twice the number of edges. Formally, the Handshaking Theorem can be stated as follows:

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Let $G = (V, E)$ be an undirected graph, where V is the set of vertices and E is the set of edges. Let $d(v)$ denote the degree of a vertex v , i.e., the number of edges incident to v . Then, we have:

$$\sum(d(v)) = 2 * |E|$$

where the sum is taken over all vertices v in V , and $|E|$ denotes the cardinality (i.e., number of elements) of the set E .

Proof: To prove the Handshaking Theorem, we can count the total number of endpoints of edges in the graph in two different ways: by summing the degrees of all vertices, and by counting the endpoints of edges.

Let's consider an undirected graph $G = (V, E)$, where V is the set of vertices and E is the set of edges. For any vertex v in V , let $d(v)$ denote the degree of v , i.e., the number of edges incident to v . Then, the total number of endpoints of edges in G is equal to the sum of the degrees of all vertices, i.e.,

$$\sum(d(v)).$$

On the other hand, each edge in E has two endpoints, so the total number of endpoints of edges in G is equal to twice the number of edges, i.e.,

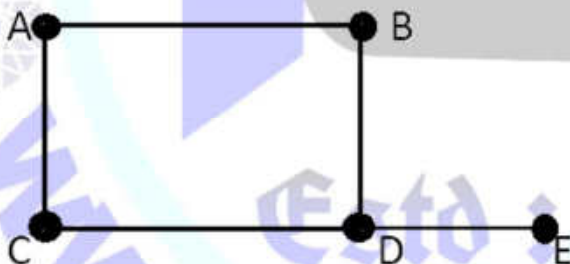
$$2 * |E|.$$

Therefore, we have:

$$\sum(d(v)) = 2 * |E|$$

which proves the Handshaking Theorem.

Example: Let's consider the following undirected graph G :



The graph has 5 vertices and 6 edges. We can calculate the degrees of each vertex as follows:

- $d(A) = 2$ (A is incident to edges AB and AC)
- $d(B) = 2$ (B is incident to edges AB and BD)
- $d(C) = 2$ (C is incident to edges AC and CD)
- $d(D) = 3$ (D is incident to edges BD, CD, and DE)
- $d(E) = 1$ (E is incident to edge DE)

Therefore, the sum of the degrees of all vertices is:

$$\sum(d(v)) = 2 + 2 + 2 + 3 + 1 = 10$$

On the other hand, the number of edges is:

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$|E| = 6$

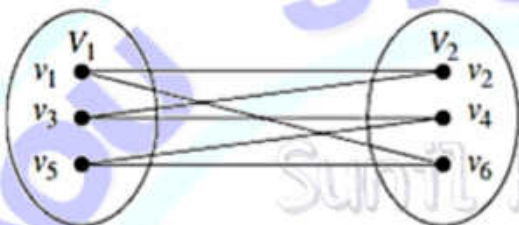
Using the Handshaking Theorem, we can verify that:

$$\sum(d(v)) = 2 * |E| \quad 10 = 2 * 5$$

which holds true.

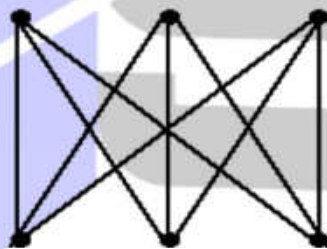
g) Show that C_6 is bipartite and K_3 is not bipartite.

Ans. C_6 is bipartite: C_6 is bipartite, because its vertex set can be partitioned into the two sets $V_1 = \{v_1, v_3, v_5\}$ and $V_2 = \{v_2, v_4, v_6\}$, and every edge of C_6 connects a vertex in V_1 and a vertex in V_2 .



K_3 is not bipartite: K_3 is not bipartite. To verify this, note that if we divide the vertex set of K_3 into two disjoint sets, one of the two sets must contain two vertices. If the graph were bipartite, these two vertices could not be connected by an edge, but in K_3 each vertex is connected to every other vertex by an edge.

$K_{3,3}$



h) Explain the terms PATH, CIRCUIT and CYCLES in context of Graphs.

Ans. In graph theory, a "graph" is a mathematical structure that consists of a set of vertices (also known as nodes) and a set of edges that connect them.

1. **PATH:** A path in a graph is a sequence of vertices connected by edges. In other words, a path is a route through the graph from one vertex to another. The length of a path is the number of edges it contains.
2. **CIRCUIT:** A circuit (also called a cycle) in a graph is a closed path that starts and ends at the same vertex. In other words, a circuit is a path that returns to its starting vertex without repeating any other vertices. The length of a circuit is the number of edges it contains.
3. **CYCLES:** A cycle is a subset of edges in a graph that form a closed path. In other words, it is a sequence of edges that connect a sequence of vertices such that the first and last vertices are the same. Cycles can be of any length, from 3 or more vertices to infinitely long paths in the case of infinite graphs.

i) What is the difference between an Eulerian graph and an Eulerian circuit?

Ans. An Eulerian graph and an Eulerian circuit are related concepts in graph theory, but they are not the same thing.

An Eulerian graph is a graph that contains an Eulerian circuit, which is a circuit that traverses every edge exactly once and starts and ends at the same vertex.

Here are the main differences between an Eulerian graph and an Eulerian circuit:

1. **Definition:** An Eulerian graph is a graph that contains an Eulerian circuit or an Eulerian path. An Eulerian circuit is a specific type of path that starts and ends at the same vertex and traverses every edge exactly once.
2. **Connectivity:** An Eulerian graph is a connected graph, which means that there is a path between any two vertices. An Eulerian circuit must also be a connected graph.
3. **Edges:** An Eulerian graph must have all of its vertices with even degree, which means that every vertex must be incident to an even number of edges. An Eulerian circuit can only exist in an Eulerian graph.
4. **Circuits vs paths:** An Eulerian circuit is a type of path that starts and ends at the same vertex and traverses every edge exactly once. An Eulerian path is a path that traverses every edge exactly once but does not necessarily start and end at the same vertex.

j) Explain the Dirac's Criterion and Ore's Criterion for Hamiltonian graphs

Ans. Dirac's criterion and Ore's criterion are two theorems that provide sufficient conditions for a graph to be Hamiltonian, which means that it contains a Hamiltonian cycle, a cycle that visits every vertex exactly once.

1. **Dirac's criterion:** Dirac's criterion states that if a graph G with n vertices has degree of each vertex at least $n/2$, then the graph is Hamiltonian. In other words, if for every vertex v in G , the degree of v is at least $n/2$, then G contains a Hamiltonian cycle.
2. **Ore's criterion:** Ore's criterion states that if a graph G with n vertices does not satisfy Dirac's criterion, but for every pair of non-adjacent vertices u and v , the sum of their degrees is at least n , then the graph is Hamiltonian. In other words, if for every pair of non-adjacent vertices u and v , the degree of u plus the degree of v is at least n , then G contains a Hamiltonian cycle.

